

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Winter 2022 - 23 Examination

Semester: 6
Subject Code: 203105395
Subject Name: Software Testing & Quality Assurance

Date: 21/10/2022
Time: 2:00pm to 4:30pm
Total Marks: 60

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions - (All are compulsory) (Each of one mark)**(15)**

1. Which of following is not part of the testing process?
 - A. Unit Testing
 - B. Requirement Analysis
 - C. Verification
 - D. Validation
2. Define a test case.
3. “ Boundary Value Analyzes the behaviour of system at maximum & minimum values”
State if the above statement is true or false.
4. Define the term Software Quality Assurance.
5. Which of following requirements forms the foundation of quality assurance?
 - A. Hardware
 - B. Software
 - C. Programmers
 - D. None of above
6. Which of following parameters shall not be considered for a SQA analysis ?
 - A. Evaluation to be performed
 - B. The magnitude of work done
 - C. Audits
 - D. All of above
7. Who is responsible for identifying, documenting & verifying the corrections made in a software?
 - A. Project Manager
 - B. Project Team
 - C. SQA team
 - D. None of them
8. A software not meeting the software requirements is the case of ____
 - A. Availability
 - B. Failure
 - C. Reliability
 - D. All of the above

9. The code of software is traversed in _____ testing.
10. "Acceptance testing is not focussed on finding the faults in a software." State whether the given statement is true or false.
11. _____ means that the product manufactured meets the design specification.
12. What does risk identification in software product mean?
13. Define Random Testing
14. What is primary role of a software tester?
15. Which software metric gives us an upper bound on number of tests needed to cover all statements of a software?

Q.2 Answer the following questions. (Attempt any three) **(15)**

- A) State four differences between verification & Validation
- B) Explain the procedure of test plan development.
- C) State and elaborate upon software risks.
- D) What is a test case? Explain how test is carried out on a software?

Q.3 A) Write about the principles of software testing? **(07)**

B) How is Glass-Box testing used to test a software **(08)**

OR

B) You want to test the functionality of a software without getting into implementation details, what testing approach will be followed? Elaborate the process. **(08)**

Q.4 A) If you want to test an individual module of a software, which testing technique will be applied? **(07)**

Explain its advantages.

OR

A) What is Software Review? Explain the types of reviews and also state importance of software review. **(07)**

B) (i) Explain the process of Fault-Based testing **(08)**

(ii) Write in brief about Scenario Testing.